

TP4

$$x \sim N(a, \sigma_x^2)$$

$$\sigma_x^2 = 2$$

$$y \sim N(b, \sigma_y^2) \text{, rge}$$

$$\sigma_y^2 = 1$$

$$H_0: a = b$$

$$H_1: a > b$$

$$x_n = \{-1.18, -6.10, 2.42\}, n = 3$$

$$y_m = \{-2.29, -2.91\}, m = 2$$

$$\alpha = 0.05$$

$$\Delta = \bar{x} - \bar{y}$$

$$\bar{x} \sim N\left(a, \frac{\sigma_x^2}{n}\right); \bar{y} \sim N\left(b, \frac{\sigma_y^2}{m}\right)$$

$$\Delta = \bar{x} - \bar{y} \sim N(a - b, \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m})$$

~~$$Z = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}}$$~~

$$Z = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}}$$

$$H_0: a = b$$

$$a - b = 0$$

$$Z \sim N(0, 1)$$

~~reject H0~~

$$H_1: a > b$$

$$\bar{x} = \frac{-1.11 - 6.10 + 2.42}{3} \approx -1.5967$$

$$\bar{y} = \frac{-2.29 - 2.91}{2} = -2.6$$

$$\tilde{\Delta} = \bar{x} - \bar{y} \approx 1.0033$$

$$S = \sqrt{\frac{s_x^2}{n} + \frac{s_y^2}{m}} = \sqrt{\frac{2}{3} + \frac{1}{2}} = \sqrt{\frac{7}{6}} \approx 1.0801$$

$$\tilde{z} = \frac{\tilde{\Delta}}{S} \approx 0.9289$$

$H_1: a > b$ - правосторонняя альтернатива.

$$z_{1-\alpha} \approx 1.6449$$

$$G = \{z > 1.6449\}$$

$$p\text{-value} = P(z > \tilde{z} | H_0) = 0.1765$$

$$p\text{-value} > \alpha, \tilde{z} < 1.6449$$

$\Rightarrow H_0$ не отвергается

$$\cdot S = a - b$$

$$S = \sqrt{\frac{s_x^2}{n} + \frac{s_y^2}{m}}$$

$$W(S) = P\left(\frac{\bar{x} - \bar{y}}{S} > z_{1-\alpha} \mid H_1\right)$$

$$W(\delta) = P\left(\frac{\bar{x} - \bar{y} - \delta}{s} > z_{1-\alpha} - \frac{\delta}{s} \mid H_0\right)$$

$$\frac{(\bar{x} - \bar{y} - \delta)}{s} \sim N(0; 1)$$

$$W(\delta) = 1 - \Phi\left(z_{1-\alpha} - \frac{\delta}{s}\right) = \Phi\left(\frac{\delta}{s} - z_{1-\alpha}\right)$$